Grade 6 Accelerated
Unit 6
Lesson 1 Practice

Name_______Answer Key____
Date______________________
Period______________________

Complete the table by rewriting the numeric expression in the form indicated. Then, find the value of each expression.

	Exponential Form	**Expanded Form**	**Value**
1.	$(3.2)^3$	$(3.2)(3.2)(3.2)$	32.768
2.	9^3	$(9)(9)(9)$	729
3.	$\left(\frac{3}{5}\right)^4$	$\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)\left(\frac{3}{5}\right)$	$\frac{81}{625}$
4.	$\left(\frac{5}{2}\right)^5$	$\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}\cdot\frac{5}{2}$	$\frac{3125}{32}$

5. What is the value of the expression $(2.8)^2$?

$(2.8)(2.8) = 7.84$

6. Which expression has the larger value, 3^6 or 6^3? Show your work or explain your reasoning.

$3\cdot3\cdot3\cdot3\cdot3\cdot3 = 729$

$6\cdot6\cdot6 = 216$

$3^6 > 6^3$ since $729 > 216$.

7. Select all expressions that are equivalent to 5^4.

- ☐ $5 + 5 + 5 + 5$
- ☐ 20
- ☑ $5 \cdot 5 \cdot 5 \cdot 5$
- ☑ 625
- ☐ $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$

8. Select all expressions that are equivalent to $\left(\frac{1}{7}\right)^3$.

- ☐ $\frac{3}{7}$
- ☐ 512
- ☐ $\frac{3}{21}$
- ☑ $\frac{1}{7} \cdot \frac{1}{7} \cdot \frac{1}{7}$
- ☑ $\frac{1}{343}$

9. What is the value of the expression $(0.08)^3$?

$(0.08)(0.08)(0.08) = 0.000512$

10. What is the value of the expression $(4.9)^2$?

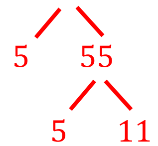
$(4.9)(4.9) = 24.01$

Grade 6 Accelerated
Unit 6
Lesson 2 Practice

Name______Answer Key_____
Date______________________
Period______________________

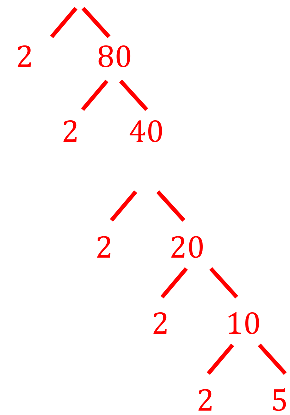
Find the prime factorization of each value.

1. 275



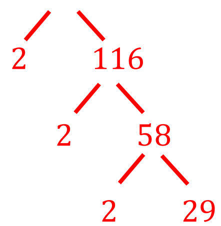
$5 \cdot 5 \cdot 11$ or $5^2 \cdot 11$

2. 160



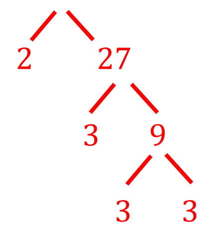
$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5$ or $2^5 \cdot 5$

3. 232



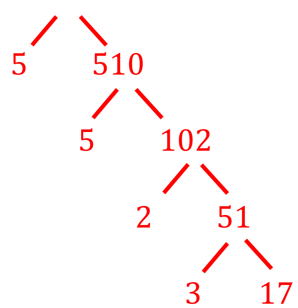
$2 \cdot 2 \cdot 2 \cdot 29$ or $2^3 \cdot 29$

4. 54



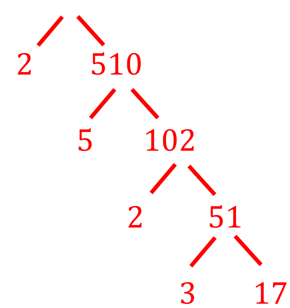
$2 \cdot 3 \cdot 3 \cdot 3$ or $2 \cdot 3^3$

5. 2550



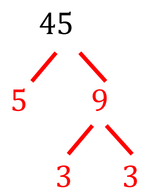
$2 \cdot 3 \cdot 5 \cdot 5 \cdot 17$ or $2 \cdot 3 \cdot 5^2 \cdot 17$

6. 1020



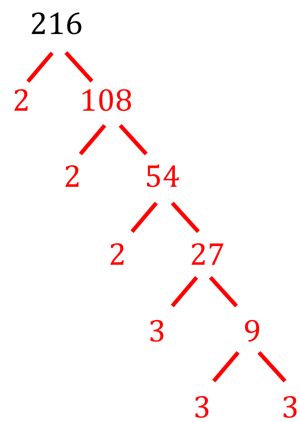
$2 \cdot 2 \cdot 3 \cdot 5 \cdot 17$ or $2^2 \cdot 3 \cdot 5 \cdot 17$

7.



$3 \cdot 3 \cdot 5$ or $3^2 \cdot 5$

8.



$2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3$ or $2^3 \cdot 3^3$

9. Find a number that meets the following conditions.
- The value of the number is less than 500.
- One of its prime factors has an exponent of 5.
- When expressed in exponential form, its prime factorization has 3 bases.

$2^5 \cdot 3 \cdot 5 = 480$

10. Find a number that meets the following conditions.
- The value of the number is less than 5000.
- One of its prime factors has an exponent of 2.
- When expressed in exponential form, its prime factorization has 5 bases.

$2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 = 4620$

Grade 6 Accelerated
Unit 6
Lesson 3 Practice

Name___Answer Key_______
Date______________________
Period_____________________

1. Which situation could be represented by the numerical expression $-8 + 12$?

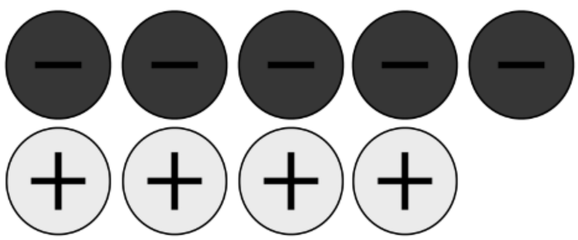
A. Jake deposited $\$4$ into his account and withdrew $\$12$.

B. Benjamin withdrew $\$8$ from his account and withdrew another $\$12$.

C. Carol deposited $\$8$ into her account and withdrew $\$10$.

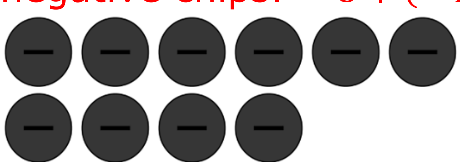
D. Debbie withdrew $\$8$ from her account and deposited $\$12$.

2. Saoirse created the sketch below to represent the expression $-5 + (-4)$.

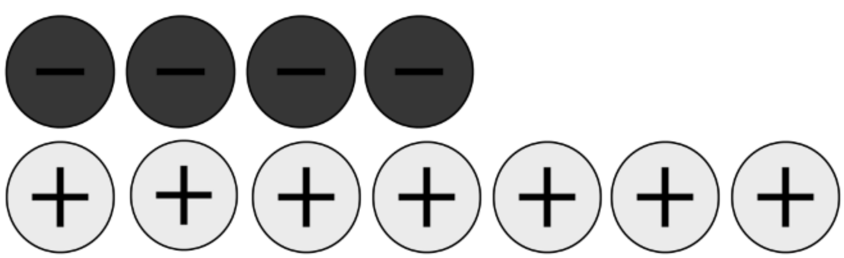


Her sketch is incorrect. Explain why and create a new sketch to find the sum.

Saoirse did not represent the expression with the correct number of negative chips. $-5 + (-4) = -9$



3. Integer chips are shown.



Which scenario could be represented by the integer chips?

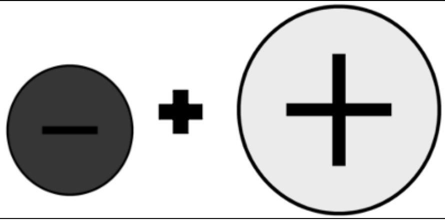
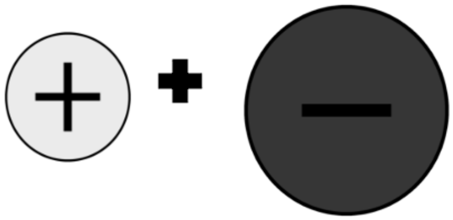
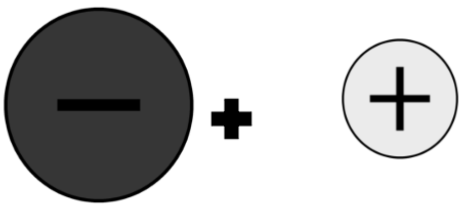
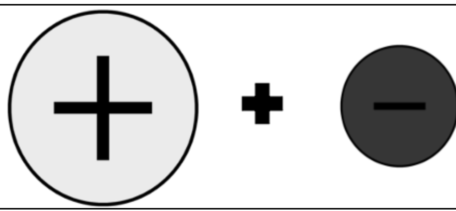
A. Octavian deposited $\$4$ into his account and withdrew $\$3$.

B. Thelonious withdrew $\$4$ from his account and withdrew another $\$7$.

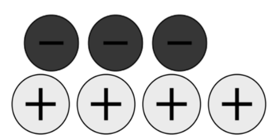
C. Lauren deposited $\$4$ into her account and withdrew $\$7$.

D. Linda withdrew $\$4$ from her account and deposited $\$7$.

Write an example of the rule. The first row has been completed.

	Visual Model of Addition	Sign of Sum	Example
			$-3+7=4$
4.			Answers will vary $3+(-7)=-4$
5.			Answers will vary $-8+2=-6$
6.			Answers will vary $8+(-2)=6$

Complete the table for the expression below.

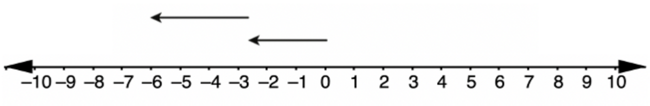
	Expression	$4+(-3)$
7.	Integer Chips	
8.	Money Sentence	Lauren has $4 in her wallet and spends $3 on lunch.
9.	Sign of the Sum	Positive (+)
10.	Sum	+1

Grade 6 Accelerated
Unit 6
Lesson 4 Practice

Name_______Answer Key____
Date________________________
Period_______________________

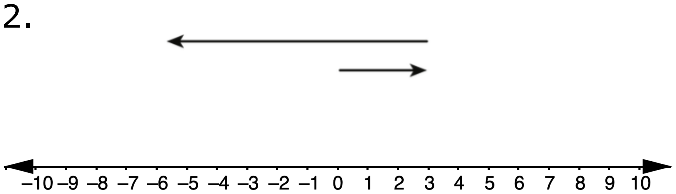
For each number line model, write the addition expression represented and find the sum.

1.



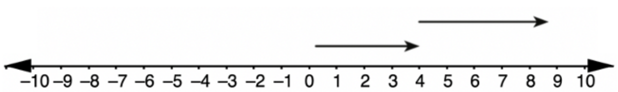
Addition Expression	Sum
$-3+(-3)$	-6

2.



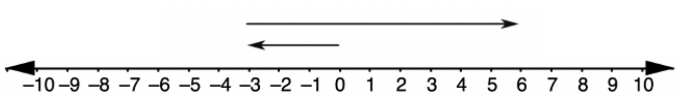
Addition Expression	Sum
$3+(-9)$	-6

3.



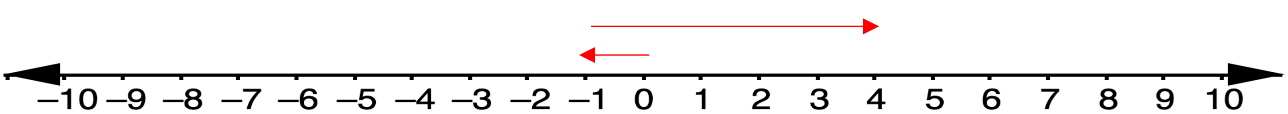
Addition Expression	Sum
$4+5$	9

4.



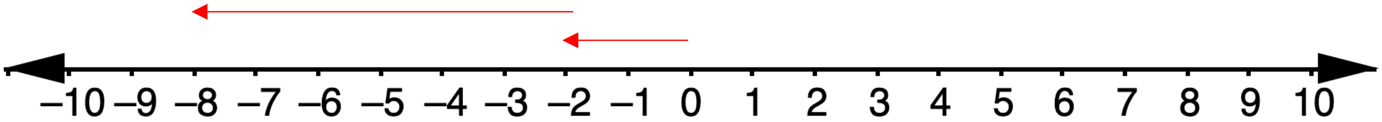
Addition Expression	Sum
$-3+9$	6

5. Use the number line to show the sum of $^{-}1+5$. Then, complete the equation.



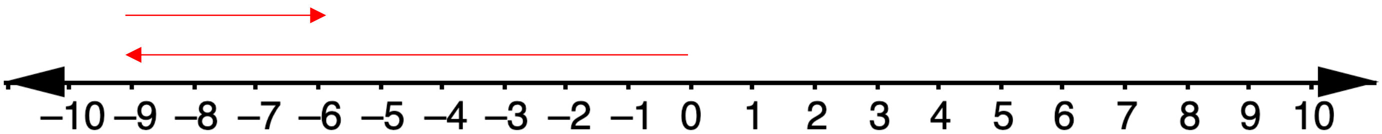
$^{-}1+5=$ __4____

6. Use the number line to show the sum of $^{-}2 + (-6)$. Then, complete the equation.



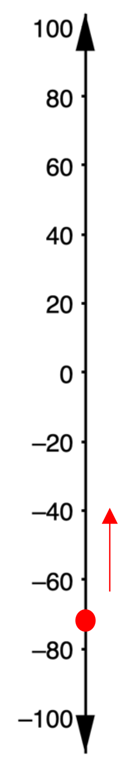
$^{-}2 + (-6) =$ ___ -8 ___

7. Use the number line to show the sum of $^{-}9 + 3$. Then, complete the equation.



$^{-}9 + 3 =$ ___ -6 ____

In a single day, the temperature in degrees Fahrenheit ($°\mathrm{F}$) of a town in Canada began at a low of $-60°\mathrm{F}$ and increased $20°\mathrm{F}$ by noon.



8. Plot a point at the low temperature.

9. Draw a ray along the number line to represent the increase of $20°\mathrm{F}$.

10. What was the temperature in the town by noon?

$-60 + 20 = -40$°F

Grade 6 Accelerated
Unit 6
Lesson 5 Practice

Name_Answer Key_________
Date______________________
Period_____________________

Find each sum.

1. $317 + (-544) = -277$

$$\begin{array}{r} 544 \\ \underline{-\ 317} \\ 227 \end{array}$$

2. $-756 + 284 = -472$

$$\begin{array}{r} 756 \\ \underline{-\ 284} \\ 472 \end{array}$$

3. $3{,}579 + (-1{,}987) = -1{,}592$

$$\begin{array}{r} 3579 \\ \underline{-\ 1987} \\ 1592 \end{array}$$

4. $-5{,}000 + (-1) = -5{,}001$

5. $-86 + (-229) = -315$

$$\begin{array}{r} 229 \\ \underline{+\ \ 86} \\ 315 \end{array}$$

6. $-8{,}352 + 4{,}686 = -3{,}666$

$$\begin{array}{r} 8352 \\ \underline{-\ 4686} \\ 3666 \end{array}$$

7. $-4{,}650 + 0 = -4{,}650$

8. $3{,}540 + (-3{,}540) = 0$

9. $-234 + 689 = 455$

$$\begin{array}{r} 689 \\ \underline{-\ 234} \\ 455 \end{array}$$

10. $-92{,}547 + 2{,}113 = -90{,}434$

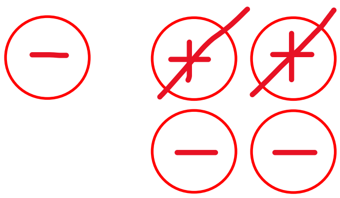
$$\begin{array}{r} 92547 \\ \underline{-\ \ 2113} \\ 90434 \end{array}$$

Grade 6 Accelerated
Unit 6
Lesson 6 Practice

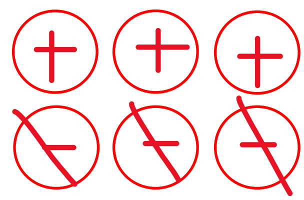
Name_______Answer Key____
Date______________________
Period_____________________

Use integer chips to find the value of each expression.

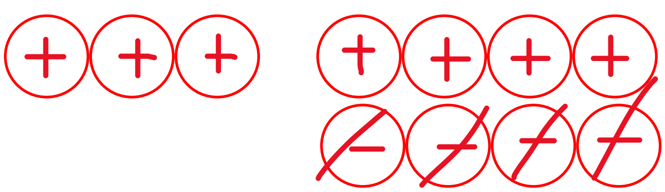
1. $-1-2=-3$



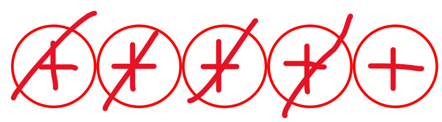
2. $0-(-3)=3$



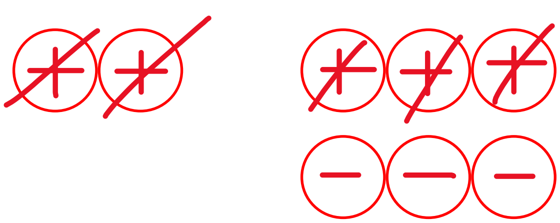
3. $3-(-4)=7$



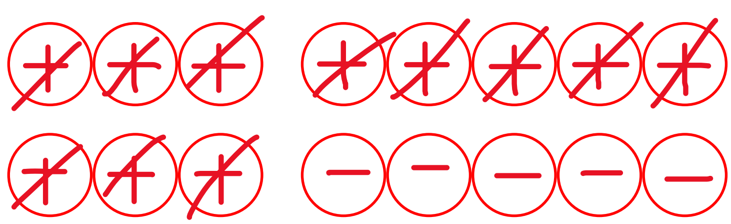
4. $5-4=1$



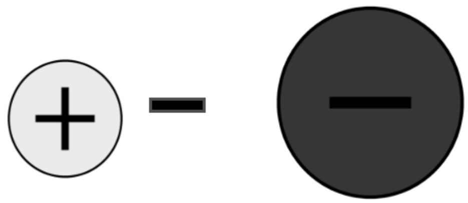
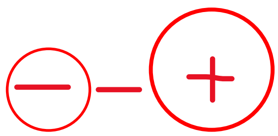
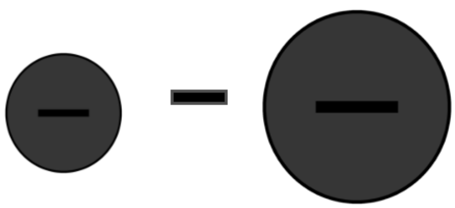
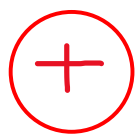
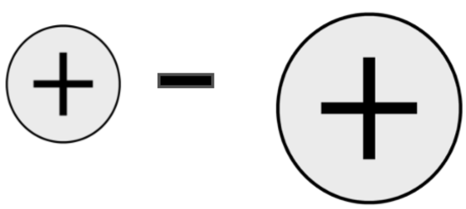
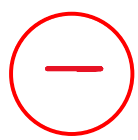
5. $2-5=-3$



6. $6-11=-5$



Complete the table.

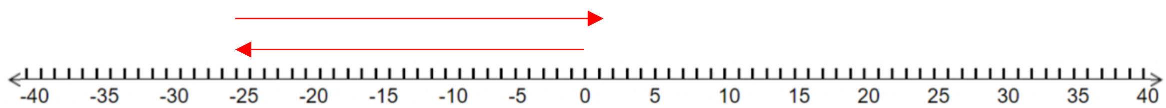
	Visual Model of Subtraction	Sign of Sum	Example
7.			Answers can vary $3-(-7)=10$
8.			$-3-6=-9$
9.			Answers can vary $-2-(-4)=2$
10.			$2-8=-6$

Grade 6 Accelerated
Unit 6
Lesson 7 Practice

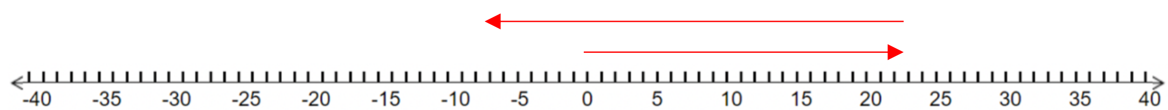
Name________Answer Key____
Date________________________
Period______________________

In questions 1-6, represent each statement on a number line and find the value.

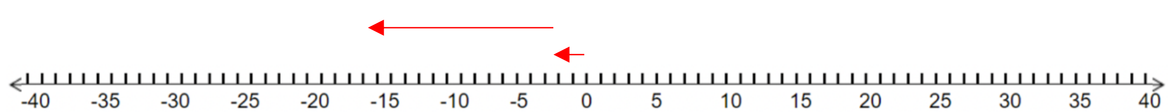
1. $-26 - (-28) = 2$



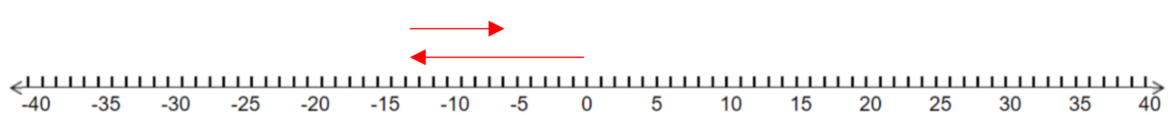
2. $23 - 30 = -7$



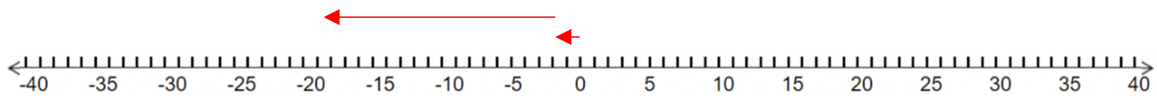
3. $-1 - 15 = -16$



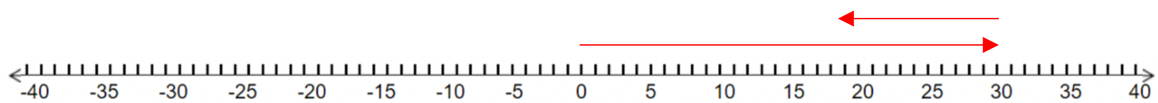
4. $(-13) - (-7) = -6$



5. $(-1) - 18 = -19$



6. $29 - 11 = 18$



In questions 7-10, rewrite each subtraction expression as an addition expression and find the value.

7. $26 - 32$

$26 + (-32) = -6$

8. $-43 - 18$

$-43 + (-18) = -61$

9. $-15 - (-43)$

$-15 + 43 = 28$

10. $(-8) - (-16)$

$-8 + 16 = 8$

Grade 6 Accelerated
Unit 6
Lesson 8 Practice

Name______Answer Key_____
Date______________________
Period____________________

Evaluate each expression.

1. $-233 - 348$

$-233 + (-348)$
-581

2. $-409 - (-176)$

$-409 + (+176)$
-233

3. $503 - 568$

$503 + (-568)$
-65

4. $355 - (-213)$

$355 + (+213)$
568

5. $622 - 545$

$622 + (-545)$
77

6. $712 - (-713)$

$712 + (+713)$
1425

7. $-188 - 356$

$-188 + (-356)$
-544

8. $419 - 455$

$419 + (-455)$
-36

9. $-366-(-402)$

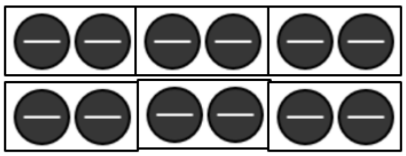
$-366+(+402)$
36

10. $-523-(-623)$

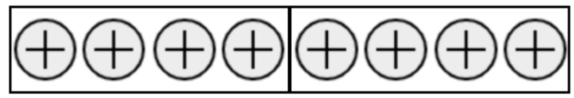
$-523+(+623)$
100

Grade 6 Accelerated — Name_Answer Key_________
Unit 6 — Date______________________
Lesson 9 Practice — Period______________________

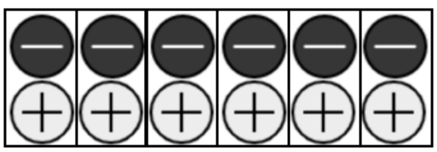
For questions 1-4, write an expression to represent the multiplication model and provide the value.

1. 

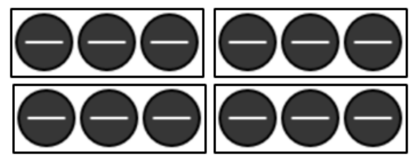
$6 \times (-2) = -12$

2. 

$4 \times 2 = 8$

3. 

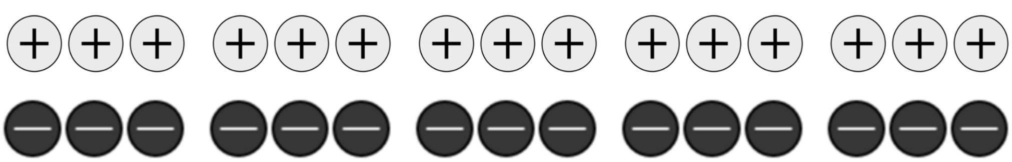
$-6 \times (-1) = 6$

4. 

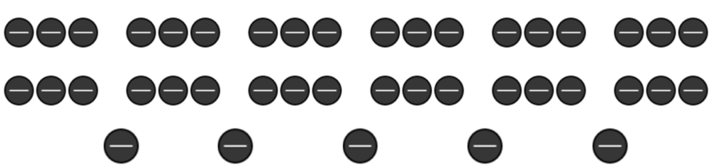
$4 \times (-3) = -12$

For questions 5-6, create a model of the expression and provide the value.

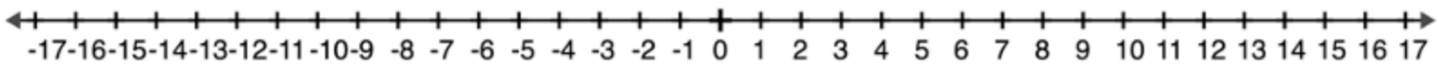
5. $(-5) \times 3$

Model	Value
	-15

6. $6(-7)$

Model	Value
	-42

For questions 7-8, use the number line to evaluate the expression.



7. $-4 \cdot 3 = -12$

8. $(-6) \times (-2) = 12$

For questions 9-10, evaluate the expression.

9. $(-11)(-9) = 99$

10. $9(-4) = -36$

Grade 6 Accelerated
Unit 6
Lesson 10 Practice

Name_______Answer Key____
Date_____________________
Period_____________________

For questions 1-6, determine the product or quotient.

1. $-28 \div 4 = -7$

2. $(56) \div (-7) = -8$

3. $\frac{-48}{-6} = 8$

4. $(-65) \div (-5) = 13$

5. $\frac{24}{-8} = -3$

6. $-48 \div 2 = -24$

For questions 7-10, create two equivalent expressions for each division expression and provide the value.

7. $(-66) \div 6 = -11$

Answers may vary

$\frac{66}{-6}$ and $-\left(\frac{66}{6}\right)$

8. $\frac{32}{-2} = -16$

Answers may vary

$\frac{-32}{2}$ and $-\left(\frac{32}{2}\right)$

9. $-60 \div (-5) = 12$

Answers may vary

$\frac{60}{5}$ and $-\left(\frac{-60}{5}\right)$

10. $64 \div (-8) = -8$

Answers may vary

$\frac{-64}{8}$ and $-\left(\frac{64}{8}\right)$

Grade 6 Accelerated
Unit 6
Lesson 11 Practice

Name______Answer Key_____
Date______________________
Period_____________________

Determine the product or quotient.

1. $-52 \div 4 = -13$

2. $(17) \cdot (-9) = -153$

$$\begin{array}{r} 17 \\ \times\ \ 9 \\ \hline 153 \end{array}$$

3. $\frac{-112}{7} = -16$

$$\begin{array}{r} 16 \\ 7\overline{)112} \\ \underline{-7} \\ 42 \end{array}$$

4. $(-85) \div (-5) = 17$

$$\begin{array}{r} 17 \\ 5\overline{)85} \\ \underline{-5\ } \\ 30 \end{array}$$

5. $(-6)(-13) = 78$

$$\begin{array}{r} 13 \\ \times\ \ 6 \\ \hline 78 \end{array}$$

6. $\frac{-111}{-3} = 37$

$$\begin{array}{r} 37 \\ 3\overline{)111} \\ \underline{-9\ } \\ 21 \end{array}$$

7. $(-6) \times (-19) = 114$

$$\begin{array}{r} 19 \\ \underline{\times \quad 6} \\ 114 \end{array}$$

8. $8 \cdot (-23) = -184$

$$\begin{array}{r} 23 \\ \underline{\times \quad 8} \\ 184 \end{array}$$

9. $(-9)(14) = -126$

$$\begin{array}{r} 14 \\ \underline{\times \quad 9} \\ 126 \end{array}$$

10. $132 \div (-4) = -33$

$$\begin{array}{r} 33 \\ 4\overline{)132} \\ \underline{-12} \\ 12 \end{array}$$

Grade 6 Accelerated
Unit 6
Lesson 12 Practice

Name______Answer Key_____
Date_______________________
Period_____________________

Determine the product or quotient.

1. $-368 \div 16 = -23$

$$\begin{array}{r} 23 \\ 16\overline{)368} \\ \underline{-32} \\ 48 \end{array}$$

2. $(48) \cdot (-18) = -864$

$$\begin{array}{r} 48 \\ \underline{\times\ 18} \\ 384 \\ \underline{+480} \\ 864 \end{array}$$

3. $\frac{-255}{17} = -15$

$$\begin{array}{r} 15 \\ 17\overline{)255} \\ \underline{-17} \\ 85 \end{array}$$

4. $(-192) \div (-8) = 24$

$$\begin{array}{r} 24 \\ 8\overline{)192} \\ \underline{-16} \\ 32 \end{array}$$

5. $(-1) \cdot (34) \cdot (-2) = 68$

6. $\frac{-224}{-4} = 56$

$$\begin{array}{r} 56 \\ 4\overline{)224} \\ \underline{-20} \\ 24 \end{array}$$

7. $(-17) \times (-18) \times (1) = 306$

$$\begin{array}{r} 17 \\ \underline{\times\ 18} \\ 136 \\ \underline{+170} \\ 306 \end{array}$$

8. $5 \cdot (-12) \cdot 6 = -360$

$$\begin{array}{r} 30 \\ \underline{\times\ 12} \\ 60 \\ \underline{+300} \\ 360 \end{array}$$

9. $(-9) \cdot (-8) \cdot (-2) = -144$

$-72(-2)$

10. $108 \div (-12) = -9$

Grade 6 Accelerated
Unit 6
Lesson 13 Practice

Name_______Answer Key____
Date______________________
Period______________________

Evaluate each expression.

1. $(3)^5 = 243$

$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$

2. $(-3)^3 = -27$

$(-3)(-3)(-3)$

3. $\left(-\frac{2}{3}\right)^4 = \frac{16}{81}$

$\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)$

4. $\left(\frac{1}{3}\right)^5 = \frac{1}{243}$

$\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)\left(\frac{1}{3}\right)$

5. The expression $(-6)^3$ will result in a [○ negative ○ positive] value because the [○ base ○ exponent] is an [○ odd ○ even] number.

For questions 6-10, select all equivalent values or expressions.

6. $(1.8)^4 = (1.8)(1.8)(1.8)(1.8)$

- ○ 7.2
- ○ $(4)(1.8)$
- ○ 10.4976
- ○ $(3.24)(3.24)$
- ○ $1.8 \cdot 1.8 \cdot 1.8 \cdot 1.8$

7. $(-3)^5$

- ○ -15
- ○ -243
- ○ $(5)(-3)$
- ○ $(9)(-3)$
- ○ $(-3)(-3)(-3)$

8. $\left(\frac{1}{4}\right)^3$ $\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{1}{64}$

- ○ $\frac{1}{12}$
- ○ $\frac{1}{64}$
- ○ $\left(\frac{1}{16}\right)(3)$
- ○ $\left(\frac{1}{16}\right)\left(\frac{1}{4}\right)$
- ○ $\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{4}$

9. $(-5)^4$

- ○ 625
- ○ $(25)^2$
- ○ -625
- ○ $(-25)(-25)$
- ○ $(-5)(-5)(-5)(-5)$

10. $(-9)(-9)(-9) = -729$

- ○ $(9)^3$
- ○ 729
- ○ $(-9)^3$
- ○ -729
- ○ $-9 \cdot (-9)^2$